



In-season estimates of ad-intact fish returning to the Krassel PIT-array

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Summary

- Current PIT-tag expansions at the Krassel PIT-array (KRS; as of July 17th)¹:
 - 655 natural-origin fish estimated from 118 observed tags and 18% sampling/tagging rate
- KRS was non-operational: 7/10, 7/11, 7/14, 7/15, and 7/16
 - Natural-origin PIT-tag passage during this period is estimated as ≤ 10 tags for an additional 56 natural-origin fish.
- The average passage date of $\geq 75\%$ of the total return across KRS is July 15th (Figure 1).
- End of season natural-origin estimate: 948
 - NOR: $(655 + 56)/.75 = 948$
- Current Lower Granite Estimate of McCall Integrated fish is 465.
- **Total ad-intact fish at KRS: 1,413**
 - $948 \text{ (NOR)} + 465 \text{ (Integrated)} = 1,413$

Assumptions

- 100% conversion of integrated fish to KRS.
- 100% detection efficiency at KRS
 - In 2021 and 2022 KRS detections efficiency was $\geq 95\%$.
- Static interpolation rate of 2 tags during the missed period at KRS when observations are typically decreasing.

¹Scripts to reproduce these summaries are available at “https://github.com/NPTfisheries/sf_salmon_harvest.com”.

site_code	rel_year	rel_group	SbyC	tag_expansion	n_tags	estimate
KRS	2020	McCall - Integrated	RAL	6.710924	1	7
KRS	2020	McCall - Segregated	RAL	50.028692	1	50
KRS	2021	McCall - Integrated	RAL	6.995739	51	357
KRS	2021	McCall - Integrated	RTR	1.000000	21	21
KRS	2021	McCall - Segregated	RAL	50.761811	45	2284
KRS	2021	McCall - Segregated	RTR	1.000000	25	25
KRS	2022	McCall - Integrated	RAL	8.312441	7	58
KRS	2022	McCall - Segregated	RAL	39.299292	8	314
KRS	2022	McCall - Segregated	RTR	1.000000	2	2
KRS	2023	LGR - NOR	-	5.555556	118	656
SFG	2020	McCall - Integrated	RAL	6.710924	1	7
SFG	2021	McCall - Integrated	RAL	6.995739	40	280
SFG	2021	McCall - Integrated	RTR	1.000000	11	11
SFG	2021	McCall - Segregated	RAL	50.761811	31	1574
SFG	2021	McCall - Segregated	RTR	1.000000	16	16
SFG	2022	McCall - Integrated	RAL	8.312441	6	50
SFG	2022	McCall - Segregated	RAL	39.299292	8	314
SFG	2022	McCall - Segregated	RTR	1.000000	4	4
SFG	2023	LGR - NOR	-	5.555556	204	1133

site_code	rel_group	n_tags	estimate
KRS	LGR - NOR	118	656
KRS	McCall - Integrated	80	443
KRS	McCall - Segregated	81	2675
SFG	LGR - NOR	204	1133
SFG	McCall - Integrated	58	348
SFG	McCall - Segregated	59	1908

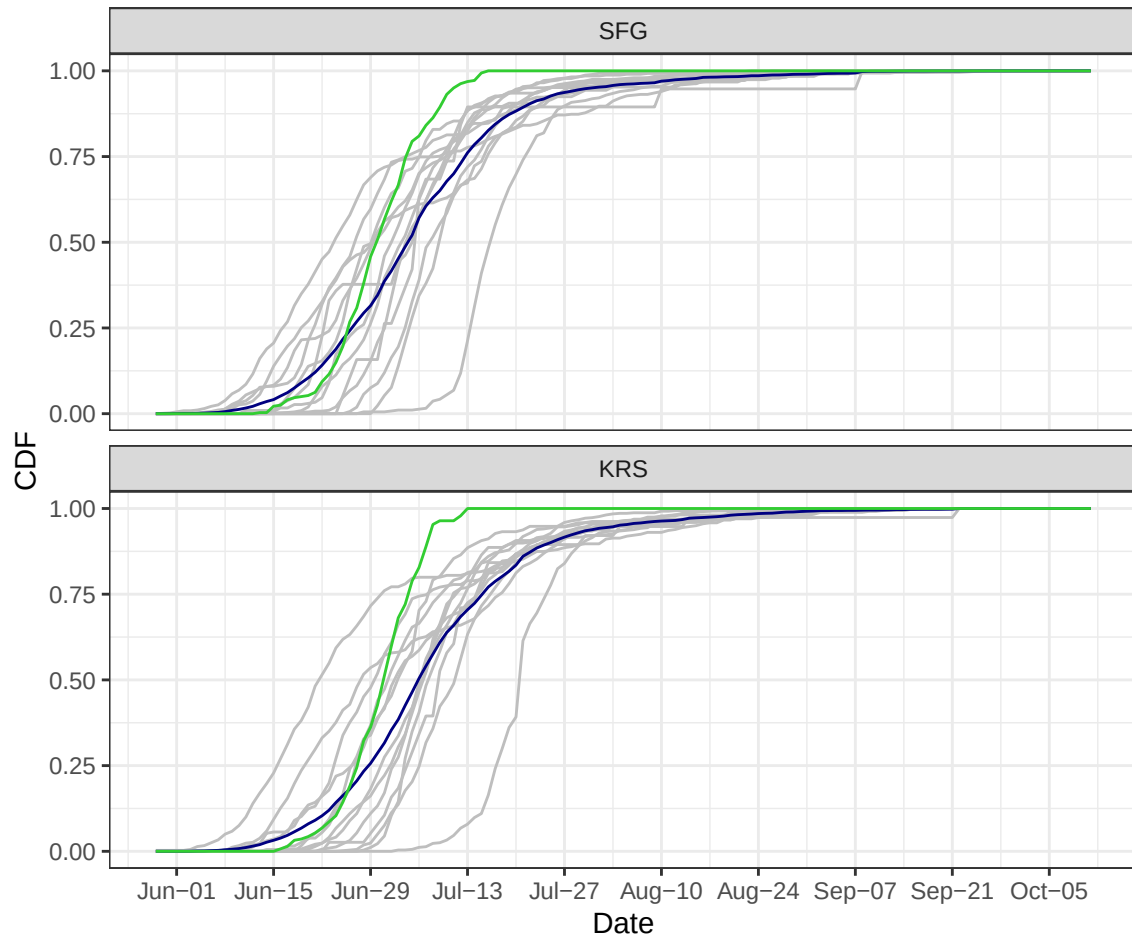


Figure 1: Run-timing of SFSR hatchery (segregated and integrated) and natural origin returns across PIT-arrays. The grey lines indicate individual spawn year returns for 2010-2022, the dark blue line is the average cumulative proportion of returns across the same years, and the green line represents the current 2023 return.